

Wenyu Chiou

Ph.D. Candidate, Civil & Environmental Engineering, Lehigh University

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AI evaluation / research engineer with catastrophe-risk domain depth.

Open to Summer 2027 internships in AI evaluation, research engineering, and risk analytics.

Work authorization: F-1 student — eligible for internships via CPT; eligible for OPT and STEM OPT extension upon graduation.

Selected Systems

- **research-hub** — Local-first research-literature pipeline whose write path is guarded by a fail-closed anti-fabrication gate: references that cannot be verified are quarantined with an explicit failure taxonomy, never silently accepted. Sole designer and maintainer end to end (architecture, authenticity gate, MCP server, CI design, evaluation suite, release process). PyPI: `research-hub-pipeline` v1.1.1; CI across 3 operating systems and 4 Python versions. Maintained open-source tool.
- **FLOODABM** — 52,141-household agent-based flood-adaptation model coupled to a catastrophe flood model with NFIP premium, payout, and deductible mechanics; calibrated on a self-collected 937-household survey (Passaic River Basin, NJ); simulated payouts validated against OpenFEMA insurance-claims data. Research prototype; archived companion code with citation metadata.
- **Cat_framework** — Team-built Hazus 6.1 seismic bridge-loss pipeline (hazard, exposure, fragility, EP/AAL); served as integrator across the pipeline; validated at three levels against the 1994 Northridge earthquake, with failures reported, not hidden. Research prototype (team capstone).
- **codex-delegate** — Cross-platform agent-delegation wrapper with anti-fabrication contracts; regression test pins a real upstream stdin hang; CI on Ubuntu and Windows. Maintained open-source tool.
- **awesome-agentic-ai-zh** — Trilingual agentic-AI curriculum with CI-enforced locale parity; 4.5k+ GitHub stars (July 2026). Maintained open-source tool.

Open-Source Contributions

- 7 merged pull requests in third-party open-source projects, including:
- `BuilderIO/agent-native` — four merged PRs, including a root-caused engine-selection bug fix with a regression test that fails on main.
- `Nanako0129/coralline` — pure-bash UTF-8 / East-Asian-width display fix.
- `ZhuLinsen/daily_stock_analysis` — Taiwan-market data feature, re-scoped through maintainer review.
- `NVIDIA/skills` — documentation fix.

Publication

- Yang, Y. C. E., & Chiou, W. (2026). Leveraging Large Language Models for Agent-Based Simulation of Human–Water System Interactions. *Water Resources Research*, 62(6), e2025WR042111. <https://doi.org/10.1029/2025WR042111> — Published; second author of two.

Education

- **Ph.D., Civil & Environmental Engineering**, Lehigh University — 2024 – present · expected 2028
- **M.S., Hydrological and Oceanic Sciences**, National Central University — Sep 2021 – Jun 2023
- **B.S., Earth Sciences**, National Central University — Sep 2017 – Jun 2021

Skills

- Programming and tooling: Python; testing and CI; Git; cross-platform (Windows/POSIX) engineering
- AI systems: evaluation design; MCP/agent tooling
- Statistical modeling: structural equation modeling (SEM); Bayesian calibration
- Risk analytics: FEMA Hazus, fragility functions, exceedance-probability curves, NFIP mechanics